



NINE MEN'S MORRIS

VARIANT LIBRARY

The ancient mill game, played on stone, cloth, and wood across five continents

7 VARIANTS	2 PLAYERS	7+ AGE
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Published by Moddable Games · v0.1.1 · Printed 22 July 2026

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Nine Men's Morris — also called Mills or Mühle — is one of the oldest known strategy games, with boards scratched into Roman stone floors, Egyptian temple rooftops, and medieval wooden cathedral furniture. The game belongs to a family called Mill games: placement and movement games in which the goal is to form three-in-a-row lines (mills) and remove opponent pieces. Variants range from the tiny Three Men's Morris (a 3×3 grid) to the elaborate Twelve Men's Morris with diagonal connections.

This library includes 7 playable variants from the Mill game family.

CLASSIC SIZES

- ♦ [Three Men's Morris](#) — The 3×3 ancestor: 3 pieces per side, no movement phase, pieces teleport
- ♦ [Six Men's Morris](#) — Two concentric squares; the medieval European standard before Nine was codified
- ♦ [Nine Men's Morris](#) — Three concentric squares; the definitive and most widely played form
- ♦ [Twelve Men's Morris](#) — Three concentric squares with added diagonal connections; 12 pieces per side

MODERN VARIANTS

- ♦ [Lasker Morris](#) — Move or place on each turn (not sequential phases); designed by World Chess Champion Emanuel Lasker

AFRICAN FORMS

- ♦ [Morabaraba](#) — South African national game: cows instead of men, additional three-piece ending rules
- ♦ [Shax](#) — Somali: capture instead of forming a mill during the placement phase



Printed 22 July 2026

V0.1.1

NINE MEN'S MORRIS · VARIANT 1 OF 7

Three Men's Morris

3×3 grid (9 intersections)

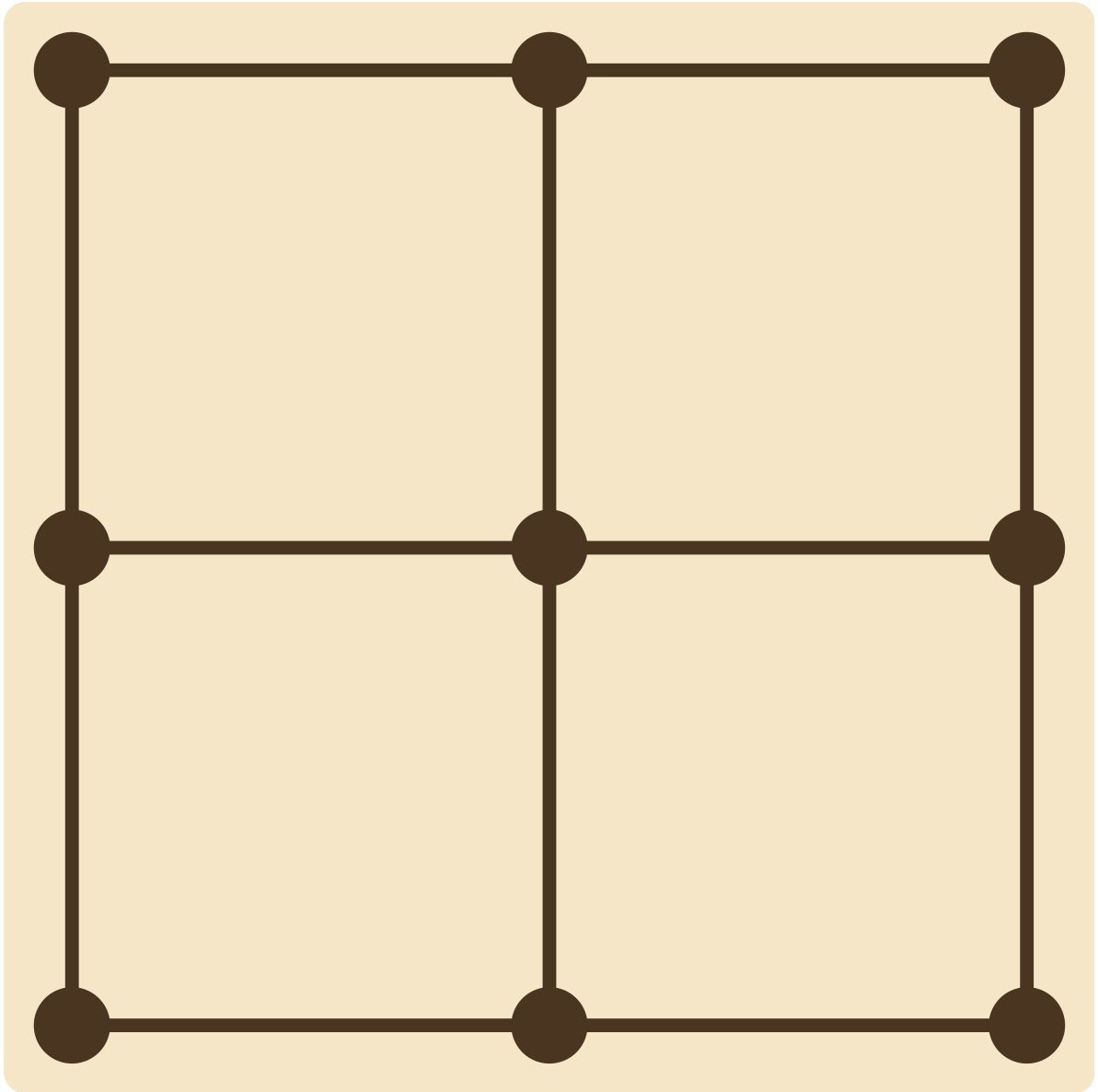
BOARD

2

PLAYERS

THREE MEN'S MORRIS

The smallest and oldest member of the Mill game family, played on a 3×3 grid with three pieces per side. Unlike the larger Morris games, Three Men's Morris has no sequential placement phase — pieces are placed and can immediately jump to any empty square on subsequent turns. The game is essentially Tic-Tac-Toe with mobile pieces, preventing draws through the ability to rearrange your position.



Three Men's Morris — board layout

<https://engine.moddable.games/play/?game=morris&variant=three-mens-morris>

ITEM	QTY	NOTES
Board	1	3×3 grid of intersections (9 positions)
Pieces	6	3 per player in contrasting colours

SETUP

The board is a 3×3 grid of points, connected horizontally, vertically, and (in most versions) diagonally. The board begins empty.

PHASE 1 — PLACEMENT

Players alternate placing one piece per turn onto any empty intersection until each player has placed all 3 pieces.

If a player forms a three-in-a-row during placement, they win immediately.

PHASE 2 — MOVEMENT

Players alternate moving one piece per turn. A piece may move to **any empty intersection** on the board — not just adjacent ones. This teleportation distinguishes Three Men's Morris from all larger variants.

The goal is to form a three-in-a-row (row, column, or diagonal, depending on board variant). On a diagonal-connected board, eight possible mills exist.

WINNING

The first player to have all three pieces in a straight line (row, column, or diagonal) wins.

DRAW PREVENTION

The free movement (jump anywhere) mechanic means that — unlike Tic-Tac-Toe — Three Men's Morris very rarely draws, because positions can always be rearranged. A draw occurs only if the same position repeats three times with the same player to move.

ATTRIBUTION

Three Men's Morris. Traditional game, Roman and medieval European origin. Public domain. Source: Murray, H.J.R. — *A History of Board-Games Other Than Chess* (1952, Oxford University Press); Wikipedia (CC-BY-SA).

NINE MEN'S MORRIS · VARIANT 2 OF 7

Six Men's Morris

Two concentric squares (16 intersections)

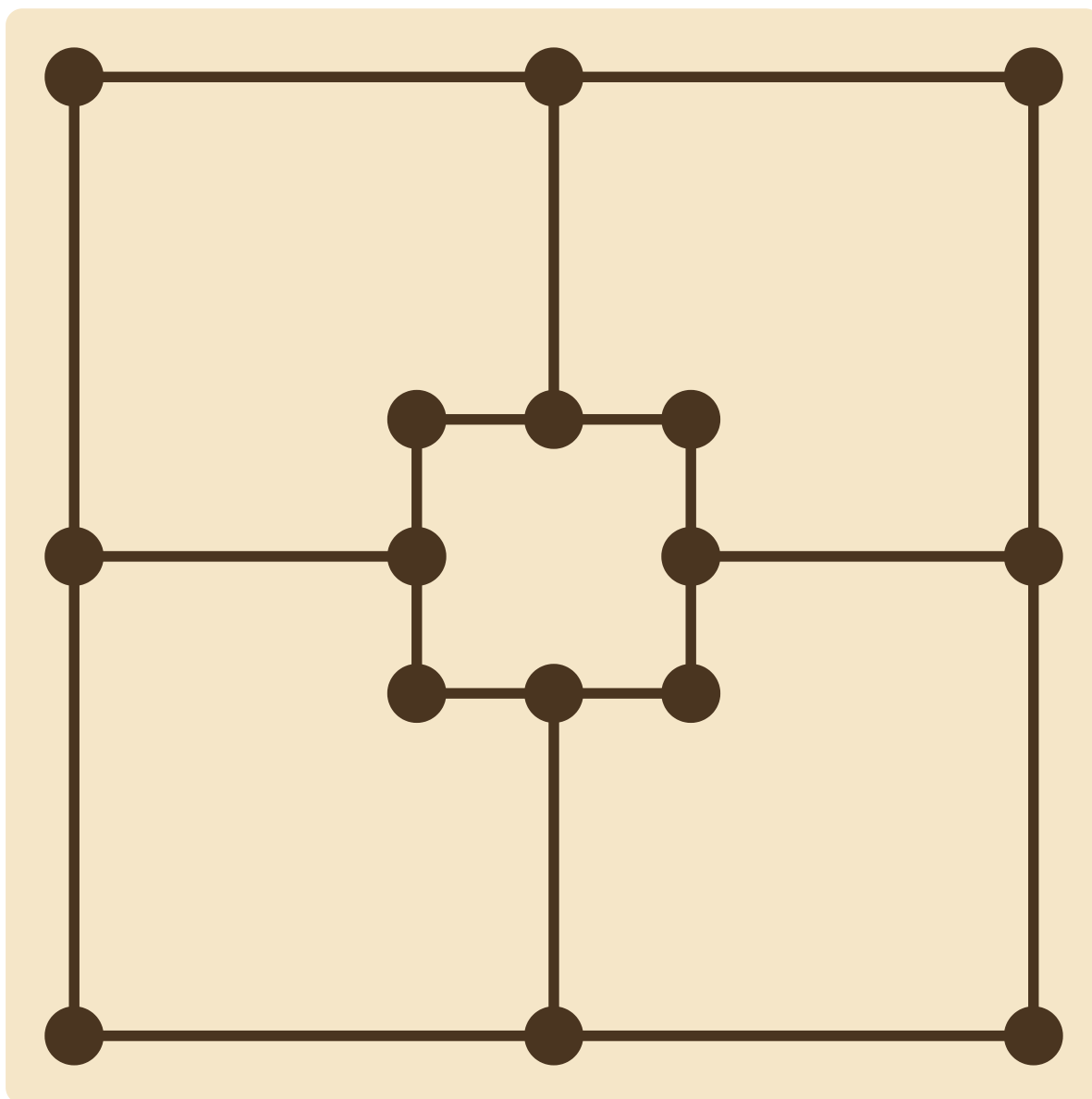
BOARD

2

PLAYERS

SIX MEN'S MORRIS

The medieval standard form of the Mill game family, widely played across Europe from the 12th to 15th centuries before Nine Men's Morris became dominant. Six Men's Morris uses two concentric squares joined at the midpoints of each side. With 6 pieces per side and 16 intersection points, the game is faster and more tactical than Nine Men's Morris, with less room for the extended blocking structures the larger game allows.



Six Men's Morris — board layout

<https://engine.moddable.games/play/?game=morris&variant=six-mens-morris>

ITEM	QTY	NOTES
Board	1	Two concentric squares; each side of each square has 3 intersections; squares connected at midpoints
Pieces	12	6 per player in contrasting colours

BOARD

The board consists of two squares — one outer, one inner — each with 8 intersection points (4 corners + 4 midpoints). The midpoint of each side of the outer square is connected to the midpoint of the corresponding side of the inner square. Total: 16 points. Mills form along each side of either square (3 points per side), giving 8 possible mills.

PHASE 1 — PLACEMENT

Players alternate placing one piece per turn onto any empty intersection until each has placed all 6 pieces.

Mill formation during placement: If a player places a piece to complete a mill (three in a row along any side of either square), they may immediately remove any one opponent piece that is not part of a mill. An opponent piece in a mill may only be removed if no non-mill pieces remain.

PHASE 2 — MOVEMENT

Players alternate moving one piece per turn to an adjacent empty intersection (along a connecting line on the board). Diagonal movement is not allowed.

Mill formation during movement: Completing a mill during a move triggers removal of one opponent piece (same rules as placement phase). A mill can be broken and reformed on subsequent turns to trigger repeated removals.

WINNING

A player wins when the opponent is reduced to 2 pieces (unable to form any mill) or when the opponent has no legal move.

ATTRIBUTION

Six Men's Morris. Traditional game, medieval Europe. Public domain. Source: Murray, H.J.R. — *A History of Board-Games Other Than Chess* (1952, Oxford University Press); Wikipedia (CC-BY-SA).

NINE MEN'S MORRIS · VARIANT 3 OF 7

Nine Men's Morris

Three concentric squares (24 intersections)

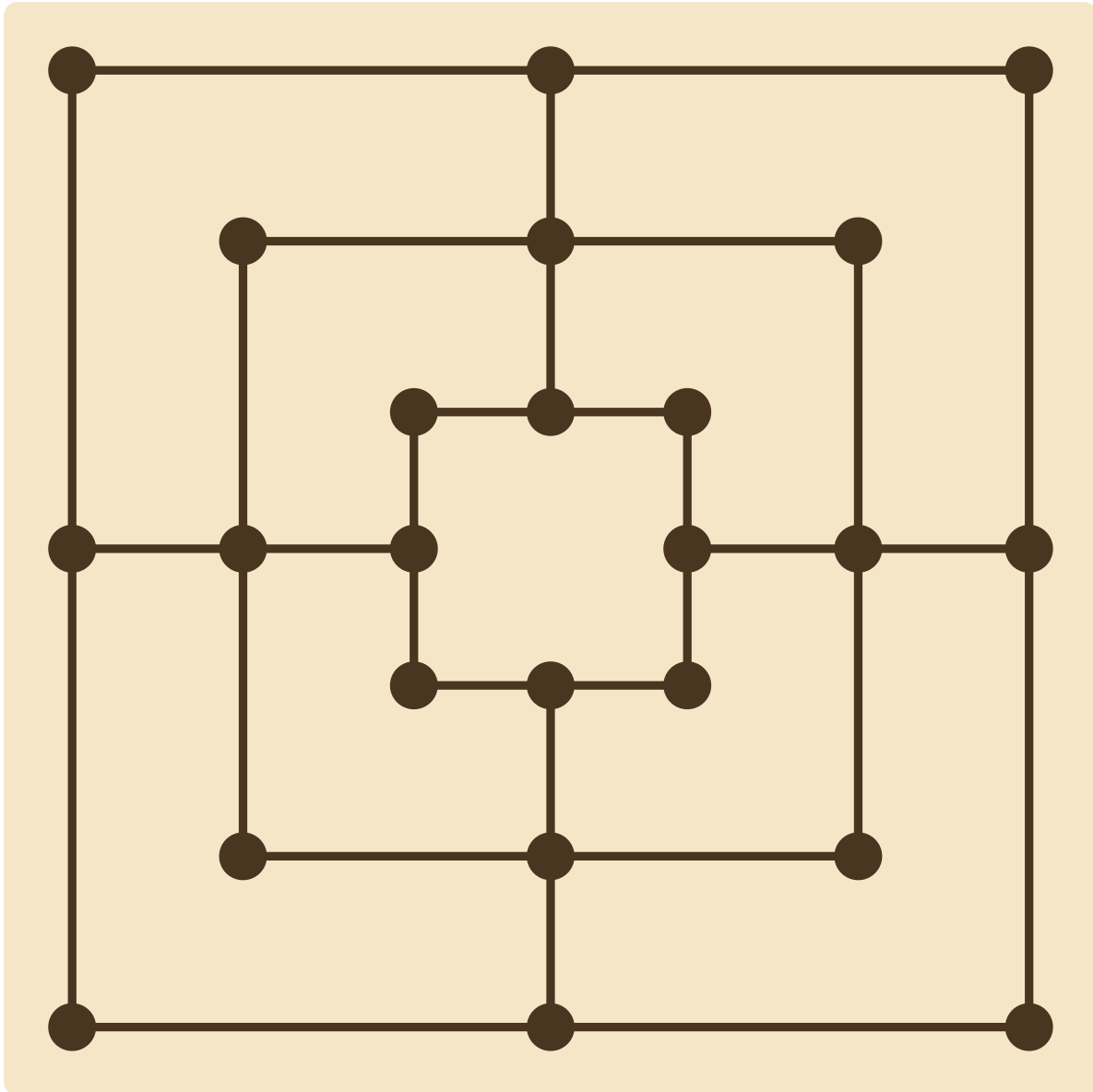
BOARD

2

PLAYERS

NINE MEN'S MORRIS

The definitive and most widely played form of the Mill game family. Nine Men's Morris uses three concentric squares joined at the midpoints of each side, creating 24 intersection points. The game has been found scratched into Roman stone at Cursor Hill and carved into medieval cathedral pews — it was ubiquitous across Europe for over 1,000 years. Today it remains popular as a pub and family game, particularly in Germany (Mühle) and Eastern Europe.



Nine Men's Morris — board layout

<https://engine.moddable.games/play/?game=morris&variant=nine-mens-morris>

COMPONENTS

ITEM	QTY	NOTES
Board	1	Three concentric squares joined at midpoints; 24 intersections
Pieces	18	9 per player in contrasting colours

BOARD

Three squares — outer, middle, inner — each with 8 intersection points (4 corners + 4 midpoints). The midpoint of each side of each square is connected to the adjacent squares' midpoints. Total: 24 points. Each side of each square forms one possible mill (3 points), giving 16 possible mills total.

PHASE 1 — PLACEMENT

Players alternate placing one piece per turn onto any empty intersection until each has placed all 9 pieces. Higher skill level: Black places first.

Mill formation during placement: Completing a mill lets you remove one opponent piece not currently in a mill. An opponent piece in a mill may only be removed if no non-mill pieces remain.

Double mills: A piece at a corner or junction can be part of two mills simultaneously. This is legal and very strong positionally.

PHASE 2 — MOVEMENT

Players alternate moving one piece to an adjacent empty intersection (along a line). No jumping or diagonal movement.

Mill and removal: Completing a mill during a move triggers removal of one opponent piece (same mill protection rules as placement).

Reopening a mill: Moving a piece out of a mill and back on subsequent turns reforms the mill and triggers another removal. This is the key middlegame tactic.

FLYING

When a player is reduced to exactly 3 pieces, they may move any piece to any empty intersection on the board — not just adjacent ones. This “flying” rule gives the weaker player a chance to escape losing positions.

WINNING

A player wins when the opponent is reduced to 2 pieces (too few to form a mill) or when the opponent has no legal move.

DRAWS

The game is a draw when both players agree, or when a position repeats three times with the same player to move.

ATTRIBUTION

Nine Men's Morris. Traditional game, Roman origin, medieval Europe. Public domain. Source: Murray, H.J.R. — *A History of Board-Games Other Than Chess* (1952, Oxford University Press); Wikipedia (CC-BY-SA).

NINE MEN'S MORRIS · VARIANT 4 OF 7

Twelve Men's Morris

**Three concentric squares with diagonals
(24 intersections)**

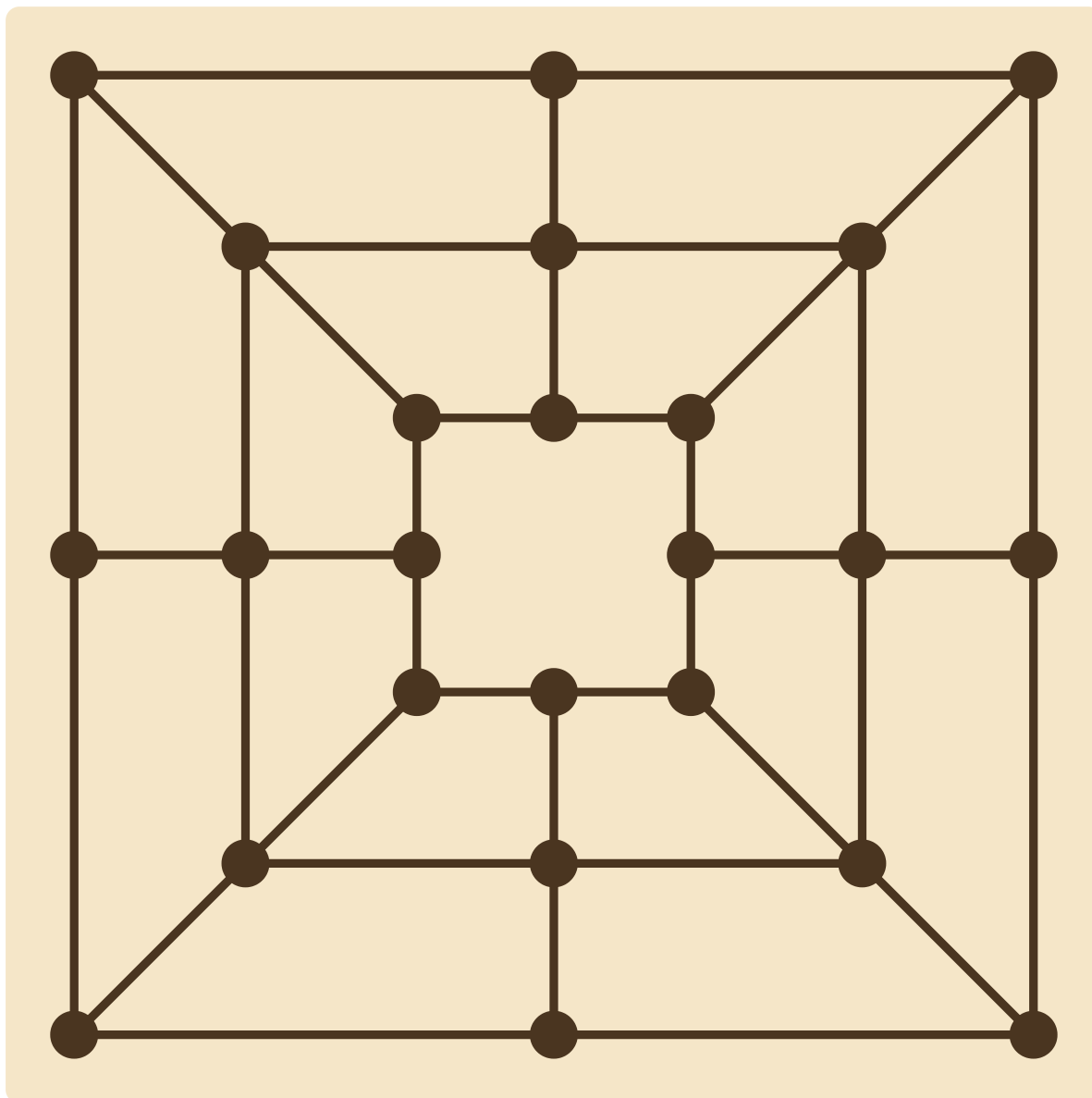
BOARD

2

PLAYERS

TWELVE MEN'S MORRIS

An extension of Nine Men's Morris using the same three concentric square board but with additional diagonal connections at every corner. The diagonals give each corner piece two possible mill directions (along the square side and along the diagonal), creating more opportunities to form mills. With 12 pieces per side (instead of 9), the game is denser and the placement phase longer. The flying rule is omitted because 12 pieces rarely reduce to 3 in normal play.



Twelve Men's Morris — board layout

<https://engine.moddable.games/play/?game=morris&variant=twelve-mens-morris>

ITEM	QTY	NOTES
Board	1	Three concentric squares with diagonals at all corners; 24 intersections
Pieces	24	12 per player in contrasting colours

BOARD

Same structure as Nine Men's Morris (three concentric squares, midpoints connected) with additional diagonal lines connecting the corners of adjacent squares. Each corner point now has three connections: along two square sides and along one diagonal. Total possible mills: 16 orthogonal (same as Nine Men's Morris) + 8 diagonal = 24 mills.

PHASE 1 — PLACEMENT

Players alternate placing one piece per turn until each has placed all 12 pieces.

Mill formation: Completing a mill (three in a row along any connected line, including diagonals) lets you remove one opponent piece not in a mill.

PHASE 2 — MOVEMENT

Players alternate moving one piece to an adjacent empty intersection (along any connection line, including diagonals).

Mill formation: Completing a mill triggers removal of one opponent piece (same rules as placement).

KEY DIFFERENCE FROM NINE MEN'S MORRIS

The diagonal connections mean that corner pieces participate in diagonal mills as well as orthogonal ones. A piece at the outer corner connects to the corresponding middle square corner, and that to the inner corner — three diagonal mills per corner axis, four axes = 4 diagonal mills (on the long diagonals from corner to corner across the board).

NO FLYING RULE

With 12 pieces per side, the endgame rarely reaches 3 pieces. Flying is not used. A player reduced to 2 pieces loses immediately (cannot form a mill).

WINNING

A player wins when the opponent is reduced to 2 pieces or cannot move.

ATTRIBUTION

Twelve Men's Morris. Traditional game. Public domain. Source: Murray, H.J.R. — *A History of Board-Games Other Than Chess* (1952, Oxford University Press); Wikipedia (CC-BY-SA).

NINE MEN'S MORRIS · VARIANT 5 OF 7

Lasker Morris

Three concentric squares (24 intersections)

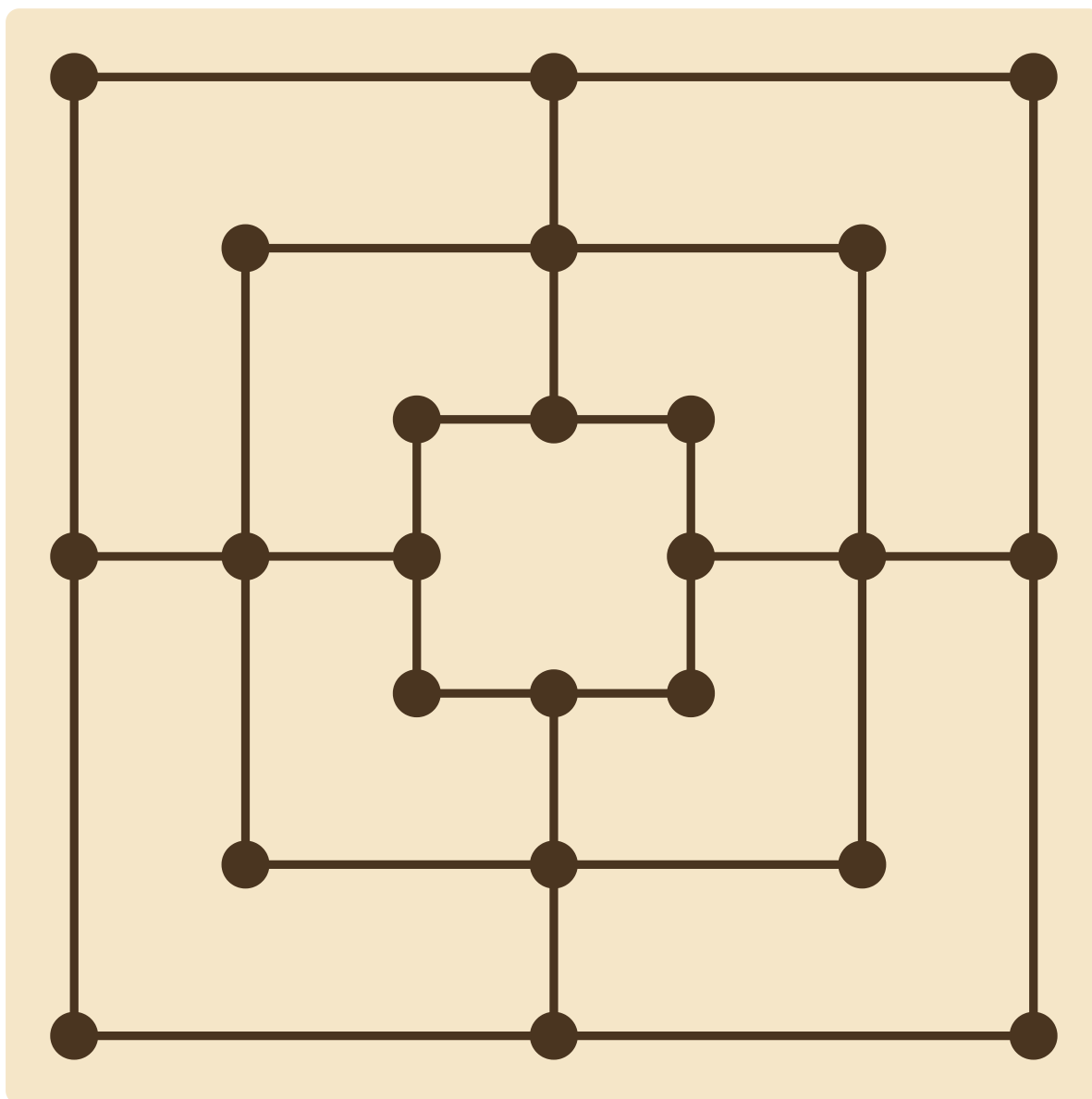
BOARD

2

PLAYERS

LASKER MORRIS

A variant of Nine Men's Morris designed by Emanuel Lasker, World Chess Champion from 1894 to 1921. Lasker modified the game to eliminate the sequential two-phase structure: instead of a placement phase followed by a movement phase, each player may on any turn either place a new piece from reserve or move an existing piece already on the board. With 10 pieces per side (one more than standard), the game creates richer interaction as both options remain available throughout most of the game.



Lasker Morris — board layout

<https://engine.moddable.games/play/?game=morris&variant=lasker-morris>

ITEM	QTY	NOTES
Board	1	Three concentric squares joined at midpoints; 24 intersections
Pieces	20	10 per player in contrasting colours

SETUP

All 20 pieces begin off the board in each player's reserve. The board starts empty.

TURNS

On each turn, a player must either:

Option A — Place: Place one piece from reserve onto any empty intersection on the board.

Option B — Move: Move one piece already on the board to an adjacent empty intersection (along a connecting line).

Both options are available on every turn until all reserve pieces have been placed. Once a player's reserve is empty, they can only move existing pieces.

MILL FORMATION

Completing a mill (three in a row along a board line) on any turn — whether by placing or moving — lets you remove one opponent piece not in a mill. The same mill protection and reopening rules as Nine Men's Morris apply.

FLYING

When a player is reduced to exactly 3 pieces, they may move any piece to any empty intersection (same as Nine Men's Morris).

WINNING

A player wins when the opponent is reduced to 2 pieces or has no legal move.

STRATEGIC NOTE

The key difference from standard Nine Men's Morris is that early placements and early movements are interleaved. A player can begin positional movement before all pieces are placed, opening more complex strategic lines.

ATTRIBUTION

Lasker Morris. Designed by Emanuel Lasker, World Chess Champion. Public domain rules. Source: Wikipedia (CC-BY-SA); Murray, H.J.R. — *A History of Board-Games Other Than Chess* (1952, Oxford University Press).

NINE MEN'S MORRIS · VARIANT 6 OF 7

Morabaraba

Three concentric squares (24 intersections)

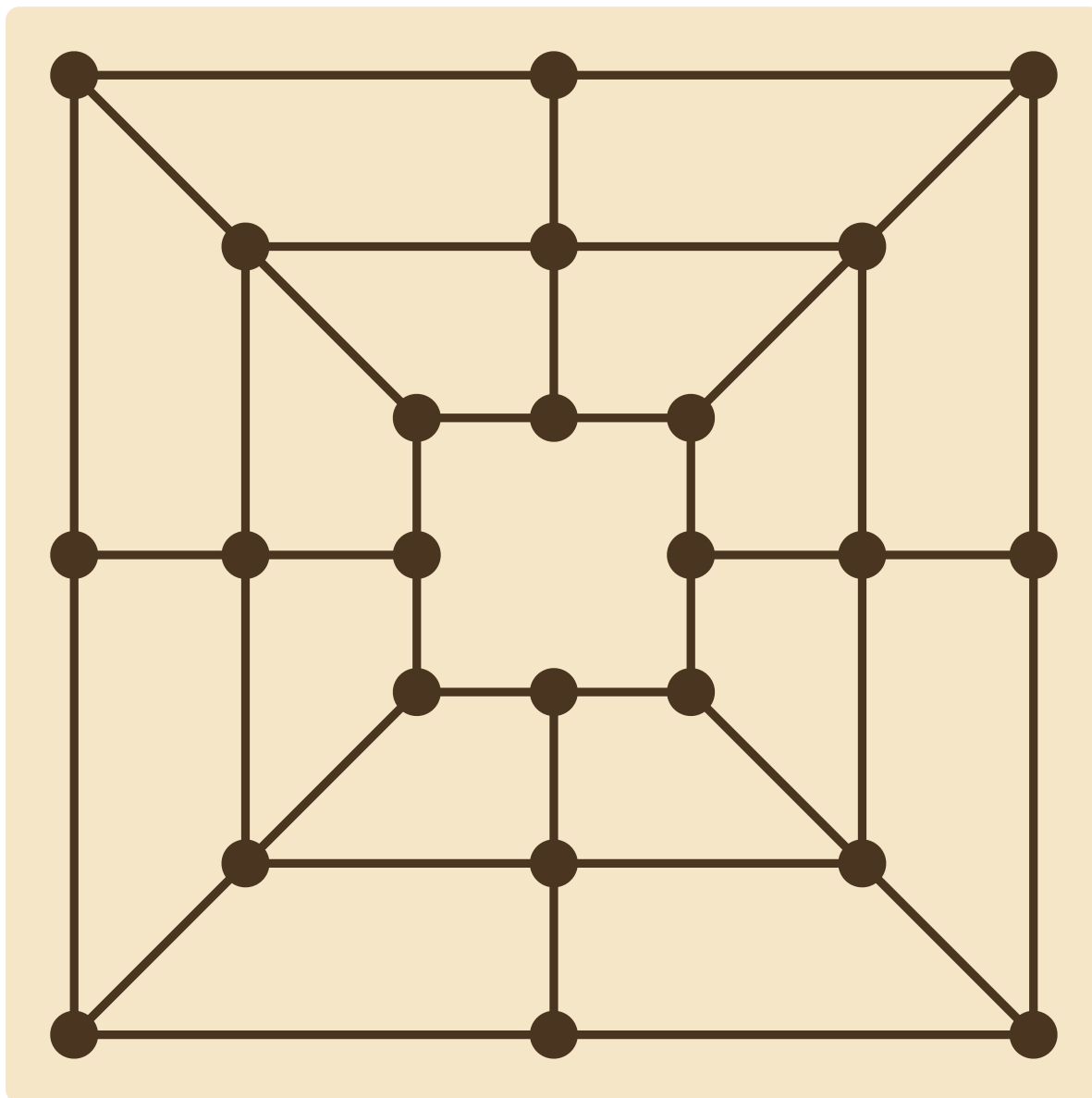
BOARD

2

PLAYERS

MORABARABA

The national board game of South Africa, played across the country and particularly popular among Zulu, Sotho, and Ndebele communities. The game is a direct descendent of Nine Men's Morris brought to southern Africa during colonial trade contact, but it has developed distinct rules and a rich local tradition. It is played as a recognised sport with national championships. Pieces are called *cows* (*inkomo*) rather than men.



Morabaraba — board layout

<https://engine.moddable.games/play/?game=morris&variant=morabaraba>

ITEM	QTY	NOTES
Board	1	Three concentric squares joined at midpoints; 24 intersections
Cows	24	12 per player in contrasting colours

PHASE 1 — PLACEMENT

Players alternate placing one cow per turn onto any empty intersection until each has placed all 12 cows. Completing a mill (three cows in a row along a board line) during placement lets you remove any one opponent cow not in a mill.

PHASE 2 — MOVEMENT

Players alternate moving one cow to an adjacent empty intersection (along a connecting line).

Mill and removal: Completing a mill during movement triggers removal of one opponent cow (same rules as placement).

Protected mills: A cow in a mill may not be captured unless all opponent cows not in mills have already been captured.

THE THREE-COW ENDGAME (FLYING)

When a player is reduced to exactly 3 cows, they may move any cow to any empty intersection on the board — not just adjacent ones. This mirrors Nine Men's Morris flying.

Special three-cow rule: When both players have 3 cows, if a player forms a mill, they may remove any opponent cow, including one in a mill. The mill protection rule is suspended in the three-cow endgame.

WINNING

A player wins when the opponent is reduced to 2 cows (cannot form a mill) or cannot move.

DRAWS

The game is a draw when a position repeats five times (more lenient than international chess draw rules), or when both players agree.

ATTRIBUTION

Morabaraba. South African national game. Public domain. Source: Mind Sports South Africa official rules; Wikipedia (CC-BY-SA).

NINE MEN'S MORRIS · VARIANT 7 OF 7

Shax

Three concentric squares (24 intersections)

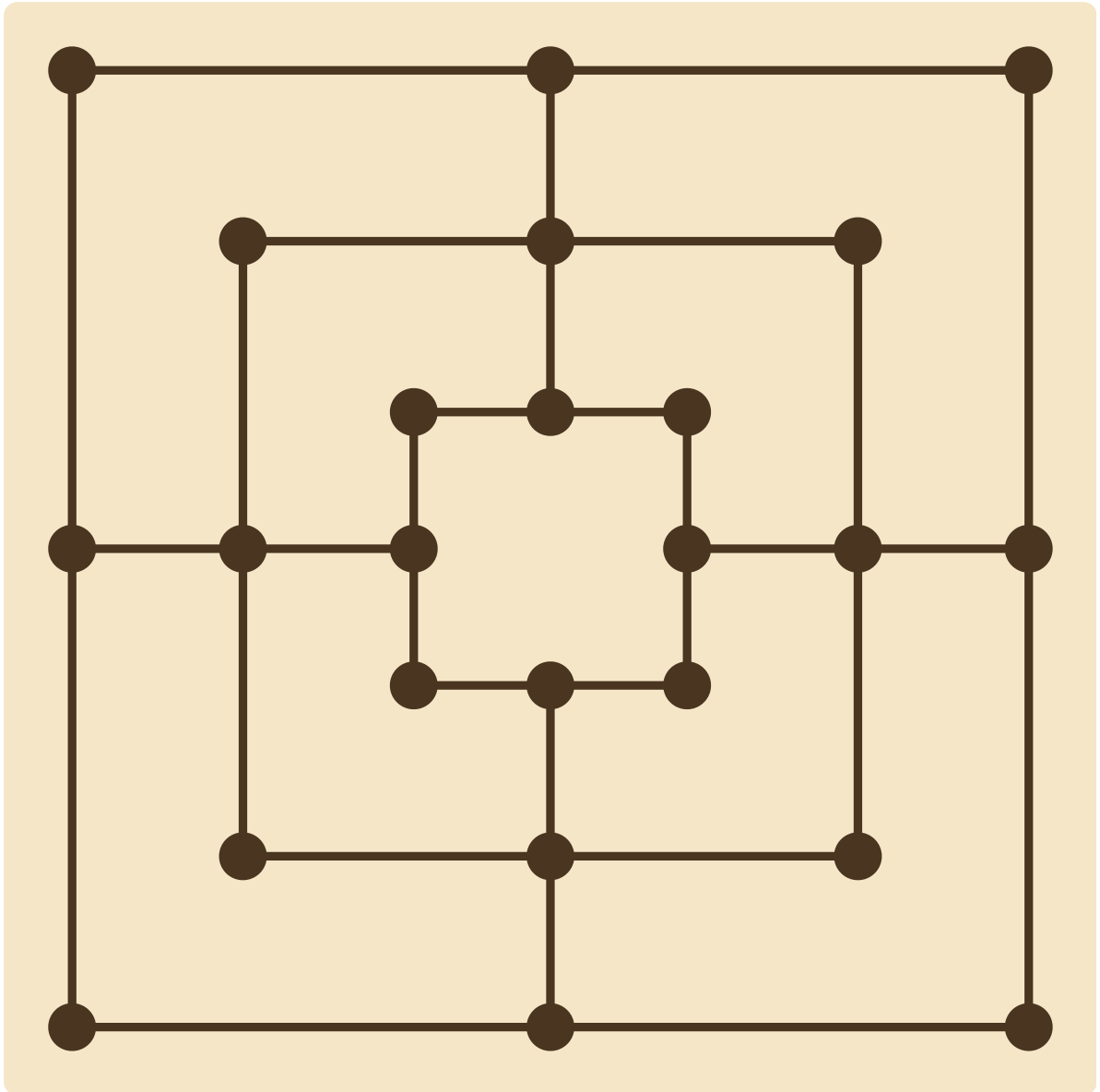
BOARD

2

PLAYERS

SHAX

The traditional board game of Somalia, played widely across the country and among the Somali diaspora. Shax uses the same three concentric square board as Nine Men's Morris but fundamentally changes the capture rule during the placement phase. In Shax, forming a mill during placement does not immediately remove an opponent piece — placement is completed peacefully by both players before any captures occur. Captures happen only during the subsequent movement phase.



Shax — board layout

<https://engine.moddable.games/play/?game=morris&variant=shax>

ITEM	QTY	NOTES
Board	1	Three concentric squares joined at midpoints; 24 intersections
Pieces	24	12 per player in contrasting colours

PHASE 1 — PLACEMENT (DHIG)

Players alternate placing one piece per turn onto any empty intersection until each has placed all 12 pieces.

Key rule: Forming a mill during the placement phase does not trigger a removal. No pieces are captured during Phase 1. Players place all pieces before any combat begins.

If the board is full after placement: If all 24 positions are occupied and no moves are possible, the player who placed last loses one piece — the piece they placed on the last turn is removed.

PHASE 2 — MOVEMENT (DHAQAAJIN) AND CAPTURE

Players alternate moving one piece to an adjacent empty intersection.

Mill formation: Completing a mill during movement lets you remove any one opponent piece, including one in a mill (unlike standard Nine Men's Morris, mill protection does not apply in Shax).

Reopening mills: Breaking a mill and reforming it on subsequent turns triggers a new capture each time.

FLYING

When reduced to 3 pieces, a player may move to any empty intersection.

WINNING

A player wins when the opponent is reduced to 2 pieces or has no legal move.

STRATEGIC NOTE

Because no captures occur during placement, both players can plan their opening positions without the immediate tactical pressure of Nine Men's Morris. The movement phase, when captures begin, is correspondingly more intense — both players arrive with full strength and begin exchanging immediately.

ATTRIBUTION

Shax. Traditional game, Somalia. Public domain. Source: Wikipedia (CC-BY-SA); academic literature on East African board games.